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**Question Paper Code: 11551**

B.E./B.TECH. DEGREE EXAMINATION, MAY 2015

Third Semester

Common to B.E – Computer Science and Engineering & B.Tech IT

**13CS303 – COMPUTER ORGANIZATION AND ARCHITECTURE**

(Regulations: MepcoR – 2013)

Duration: 3 Hours Maximum: 100 Marks

**Answer ALL Questions**

**PART – A (10 x 2 = 20 Marks)**

1. Compare and contrast RISC and CISC approaches of modern computers.
2. Express the unnormalized value 0.00011010 . . . × $2^9$ in the IEEE single-precision normalized format.
3. What are the five-step sequence of actions needed to fetch and execute instructions in a RISC-style instruction set?
4. Consider the login expression for the Counter-enable signal:

   Counter_enable = $\overline{WMFC}$ + MFC

   Why is it necessary to include both WMFC and MFC signals?
5. What do you mean by hazard in pipelining? Give an example.
6. Is it possible to reduce branch penalty in the pipelined execution of a sequence of instructions? Justify.
7. Differentiate between static and dynamic RAM.
8. Define direct memory access (DMA).
9. What is interrupt latency?
10. Give a brief account on SAS interface standard.

**PART – B (5 × 16 = 80 Marks)**

11. (a) i. Explain the various addressing modes by giving suitable examples. (10 Marks)

11. (a) ii. Perform the signed number multiplication using Booth's algorithm:

    -21 x 24 (6 Marks)

**OR**

11. (b) i. Consider the two decimal floating-point numbers I = 2.6395 x $10^2$ and J = 3.7215 x $10^4$. Explain in detail, how to perform the operations I + J and I x J. (10 Marks)

11. (b) ii. Perform the operation A ÷ B on the 5-bit unsigned number A = 10101 and B = 00101, using non-restoring division. (6 Marks)

12. (a) i. Describe in detail the datapath of a processor by identifying the five-stage structure and various registers between stages. (10 Marks)

12. (a) ii. Identify the various sequence of actions needed to fetch and execute the instruction: (6 Marks)

Store R8, X(R9)

**OR**

12. (b) i. Explain in detail how control signals are generated in RISC-style processors. (10 Marks)

12. (b) ii. Identify the various sequence of actions needed to fetch and execute the instruction:

And X(R4), R3 (6 Marks)

13. (a) i. By defining what data dependency is, explain how it is handled in hardware. (10 Marks)

13. (a) ii. Consider the following instructions at the given addresses in the memory:

| | | |
|---|---|---|
| 1000 | Add | R3, R2, #20 |
| 1004 | Subtract | R5, R4, #3 |
| 1008 | And | R6, R3, #0x3A |
| 1012 | Add | R7, R2, R4 |

Initially, registers R2 and R4 contain 2000 and 50, respectively. These instructions are executed in a computer that has a five-stage pipeline. The first instruction is fetched in clock cycle 1, and the remaining instructions are fetched in successive cycles. Assume that the pipeline provides forwarding paths to the ALU from registers RY and RZ and that the processor uses forwarding of operands.

(*a*) Draw a diagram that represents the flow of the instructions through the pipeline. Describe the operation being performed by each pipeline stage during clock cycles 1 through 8.

(*b*) Describe the contents of registers IR, PC, RA, RB, RY, and RZ in the pipeline during cycles 2 to 8. (6 Marks)

**OR**

13. (b) i. Examine the effect of conditional branch instruction and identify various techniques that can be used to mitigate the effects on execution time. (10 Marks)

13. (b) ii. Assume that 25 percent of the dynamic count of the instructions executed for a program is branch instructions. There are no pipeline stalls due to data dependencies. Static branch prediction is used with a not-taken assumption.

(A) Determine the execution times for two cases: when 35 percent of the branches are taken, and when 65 percent of the branches are taken.

(B) Determine the speedup for one case relative to the other. Express the speedup as a percentage relative to 1. (6 Marks)

14. (a) i. Describe a structure for the organization of a 16M x 32 memory module using 1M x 4 static memory chips. (10 Marks)

14. (a) ii. Explain the various kinds of ROM memories. (6 Marks)

**OR**

14. (b) i. A block-set-associative cache consists of a total of 64 blocks, divided into 4-block sets. The main memory contains 4096 blocks, each consisting of 32 words. Assuming a 32-bit byte-addressable address space, how many bits are there in each of the Tag, Set, and Word fields? Compare the same technique with the direct mapping technique. (10 Marks)

14. (b) ii. Explain the virtual-memory address translation scheme with a neat sketch. (6 Marks)

15. (a) i. Consider the situation where "device X may request an interrupt while an interrupt caused by device Y is being serviced, or several devices may request interrupts at exactly the same time". How this situation is handled by the processor. (8 Marks)

15. (a) ii. Describe an interface circuit for an 8-bit input port that can be used for connecting a simple input device, keyboard. Assume that this interface circuit is connected to a 32-bit processor that uses memory-mapped I/O and the asynchronous bus protocol. (8 Marks)

**OR**

15. (b) i. Explain the asynchronous bus operation for an input operation and identify the various sequence of events. (10 Marks)

15. (b) ii. Illustrate in detail, the sequence of events that take place during a read operation of SCSI bus standard, when used with a disk drive. (6 Marks)